North Carolina State University
ECE/MAE 535: Design of Electromechanical Systems

Design Project

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Electro-Permanent Magnetic Clamp for
Workpiece Holding During Machining

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Abstract

The process of securing the workpiece can be the most critical aspect of the machining process. This report presents the comprehensive design, analysis, and performance evaluation of an electro-permanent magnetic clamp (EPMC) tailored for secure workpiece holding during machining operations. The primary objective of this project was to develop a robust and efficient clamping system given certain design requirements. The design consists of three NdFeB magnets and two AlNiCo magnets around two poles. The design demonstrates the "on" and "off" condition of the EPMC. To get the desired results, the process began with Magnetic Circuit Analysis (MCA) of the clamp to achieve the force equations, followed by the 2D FEMM simulation of the design. OctaveFEMM was used to simulate and analyze the design. Parallely, a brute force approach was used to get the dimensions of the optimum design satisfying the holder to provide at least 500 lb of vertical reluctance force, through a factor safety of 3.0 (1500 lb total). This optimal design achieved through brute-force approach was then used to verify and optimize the values achieved through the OctaveFEMM simulation. Finally, with the optimized dimensions, a model was constructed in Ansys Workbench to examine temperature properties of Coil and permanent magnet (PM). This aspect is vital to prevent workpiece distortion due to heat. Ultimately, the design provided a highly cost-effective alternative to the traditional clamping method.

Introduction

Traditional methods of securing ferromagnetic workpiece during manufacturing processes (milling, grinding, drilling, etc.) often involve time-consuming setup procedures using clamping devices. However, the drawbacks associated with conventional clamps, such as setup time and labor intensity, have prompted the exploration of alternative solutions. One such emerging alternative utilizes reluctances force secure workpieces, significantly reducing setup times. While initial attempts involved the use of electromagnets for this purpose, challenges arose due to the need of constant electrical power, which not only generated heat but also risked dimensional changes in the workpiece. Moreover, the potential consequences of power loss during high-speed machining operations were catastrophic. To address these limitations, recent developments have led to the introduction of electro-permanent magnetic circuits, offering the advantages of reluctance force holding without the requirements for continuous electrical power. Notably, the electro-permanent magnetic clamp (EPMC) integrates both low-coercivity and high-coercivity permanent magnet materials, namely AlNiCo and NdFeB, respectively. This report delves into the design and functionality of the EPMC, elucidating its unique capabilities and potential benefits for machining operations.

Magnetic Circuit Analysis

We implemented the concepts of MAE/ECE-535 and performed the MCA analysis to try and get the forces for the EPMC. This was done utilizing matlab and manual calculations to get the values of the forces for the circuit and setup. The setup can be simulated as a circuit consisting of sets of resistances, fluxes and currents that can be also represented by the Hopkinson law of forces. Figure 1 shows the reluctances and the flux going through each of the elements. In this system we have the reluctances of the steel, air and the NdFeB and Alnico magnets.

The direction of the fluxes are based on the problem statement. The idea is to do analysis of these circuits Kirchhoff's Voltage Law and Kirchhoff's Current Law which state that the voltage of a loop should be equal to zero and the other is the sum of all current entering or leaving a node equal to zero. By using this method and applying Hopkinsons Law we could obtain the forces and flux node equations shown in Appendix B.

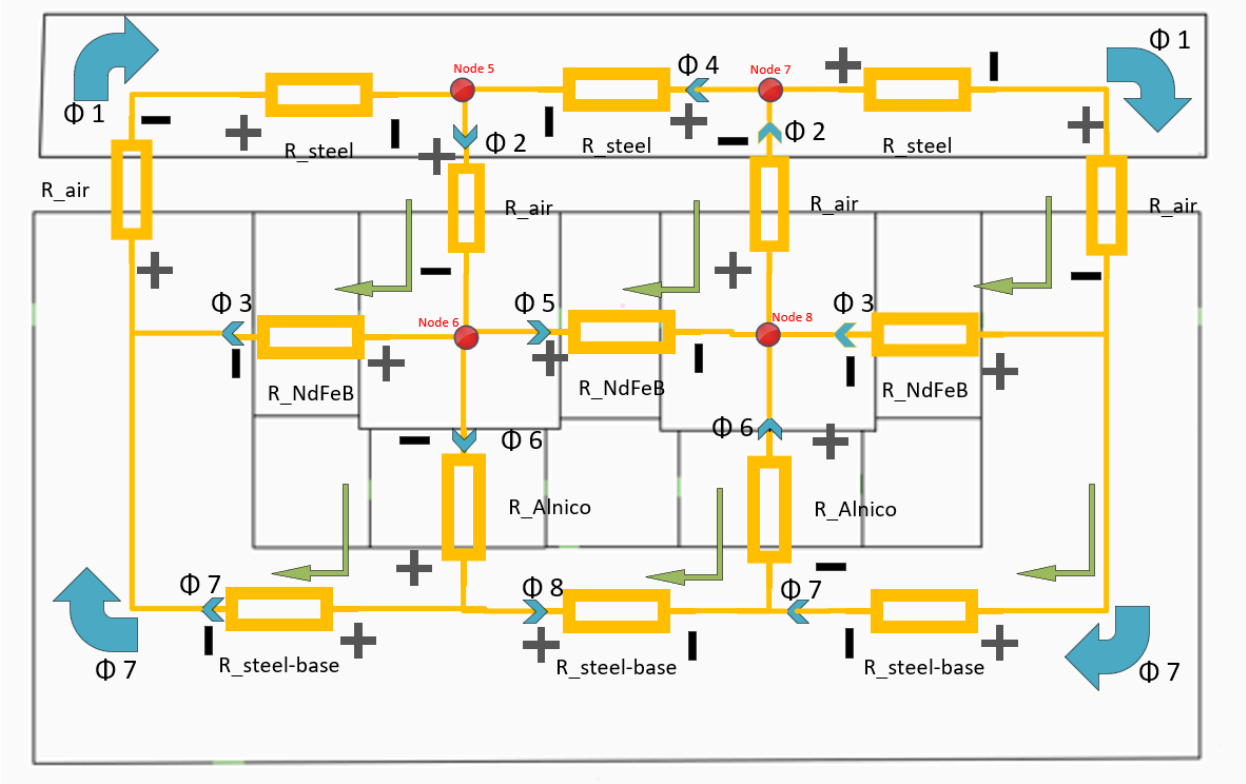


Figure 1 : Circuit represented as reluctances and fluxes

There are six loops that can be used to calculate the forces and 10 node equations that can be created. By symmetry of the system the right and left side are the same; this makes the system to have opposing forces that cancel and that can make some nodes and loops give zero. This approach is not really necessary, since we can take one side and the center and mirror it to the other side since what we have to simulate is just a part of a bigger system or piece. The chosen loop where the 2 of the center, the two on the left and the nodes are the one marked as red dots in Figure 1.

Each loop and node gives one equation that depends on the other variables. This creates a system of 8 equations, 4 for loops and nodes respectively. These equations are presented in Table 1 below. This equation was then programmed into matlab to obtain the final result. After the results were obtained we

discovered that they were not as expected. It seems that the linear approach was erroneous and did not consider several factors.

B-H Curve (slopes)			
Air (polynomial fitting)	Alnico (polynomial fitting)	NdFeB (polynomial fitting)	Steel (exponential fitting)
$B_{air} = \mu_0 H$	$B_{Alnico} = 3.6x10^{-6}H - 1.08$	$B_{NdFeB} = 1.68x10^{-6}H - .55$	$H_{steel} = .9302e^{(4.99B)}$
MMF Equations			
Air	$F_{air} = \frac{l_{air}}{\mu_0 A_{air}}$		
Alnico	$F_{Alnico} = 230,400\frac{\Phi\ l_{Alnico}}{A_{Alnico}} - 1,130,000(l_{Alnico})$		
NdFeB	$F_{NdFeB} = 230,400\frac{\Phi\ l_{NdFeB}}{A_{NdFeB}} - 296,663(l_{NdFeB})$		
Steel	$F_{Steel} = .9302e^{(4.99*\frac{\Phi}{A_{NdFeB}})}(l_{NdFeB})$		

Table 1 : Table of system of equations derived using MCA and Matlab

Working with the MCA raised some concerns after deriving equations shown in Table 1 and obtaining the results were not satisfactory. We decided that the many assumptions made in the derivation of the equations had several limitations while linearizing the system and also further added to the complexity. Had this been an iterative process and that the dimensions were to be changed many times to get an optimal system, it would have been tedious and time consuming given a system of numerous equations. Therefore, we decided to pursue an analytical approach using simulations in FEMM. The next two sections elaborate on the use of FEMM(OctaveFEMM with MATLAB) and Brute-force approaches to automate the simulation.

Brute-Force Approach

We started with a preliminary design based on the project requirements as shown in Figure 2. This design served as the foundation for a manual optimization process using trial and error. We adjusted various factors like dimensions and material grades. A 0.3mm gap between the workpiece and the EPMC was necessary to avoid direct contact.

Following were the variations and the corresponding outcomes:

A. **Case Dimensions:** To prevent flux leakage from the bottom and sides, we increased the case height to over 60 mm and the width to a total of 180mm by expanding both sides equally. 1020 steel[14] was used instead of 1025 due to the better availability of its magnetic and physical properties, and also because 1025 steel was unavailable in the FEMM 'materials' library.

B. **Steel Pole Dimensions**: The pole height was gradually extended to a maximum of 20mm, keeping the width at 50mm. This helped reduce outgoing flux towards the workpiece during the OFF-state.

C. **AlNiCo Dimensions**: We minimized height adjustments for this magnet as a grade increase alone achieved the required ON-state force (1501 lbf).

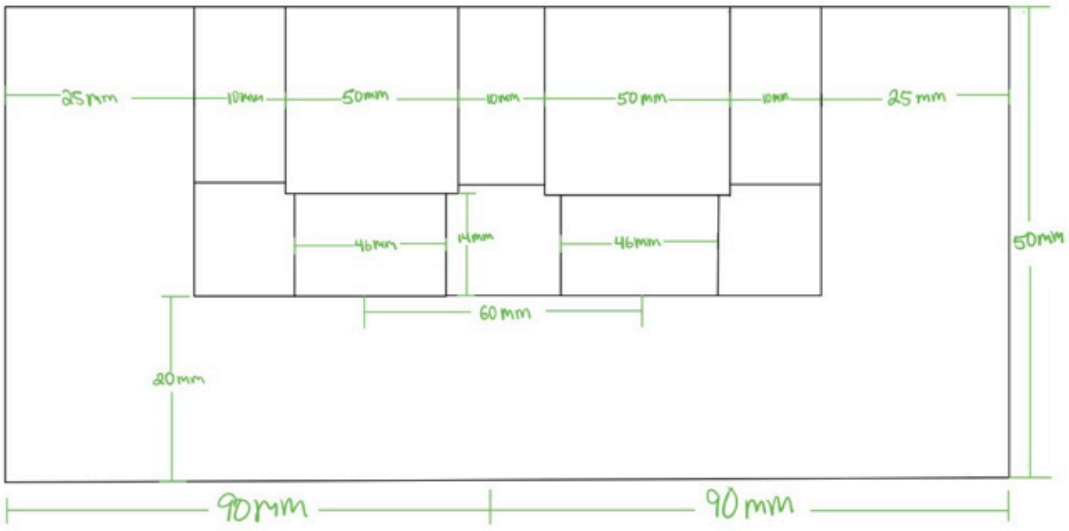


Figure 2 : Initial tentative design with given dimension requirements

D. **NdFeB Dimensions:** To maintain consistency with the other magnets, a minimum width of 10mm was maintained. Aligning the magnet's top and bottom edges with the steel pole, we found grade N33 to be optimal. This minimized flux entering the workpiece in the OFF-state (reducing force to around 10lbs) and effectively captured the field within the EPMC. In the ON-state, it provided better field distribution around the workpiece, potentially improving stability and increasing force.

E. **Magnet Grades:** Initially, the grade for AlNiCo magnets was kept the highest (LNG44) and that for NdFeB was the lowest(N30), based on the availability in FEMM. As the simulation efforts proceeded, these grades were varied and the optimal combination was found out to be N33 and LNG 40, since having the AlNiCO magnets are relatively cheap, it is affordable to have superior grades as compared to NdFeB. Grades of AlNiCo considered were LNG(44/40/37) and for NdFeB they were N(30/33/35). [2][3][5]

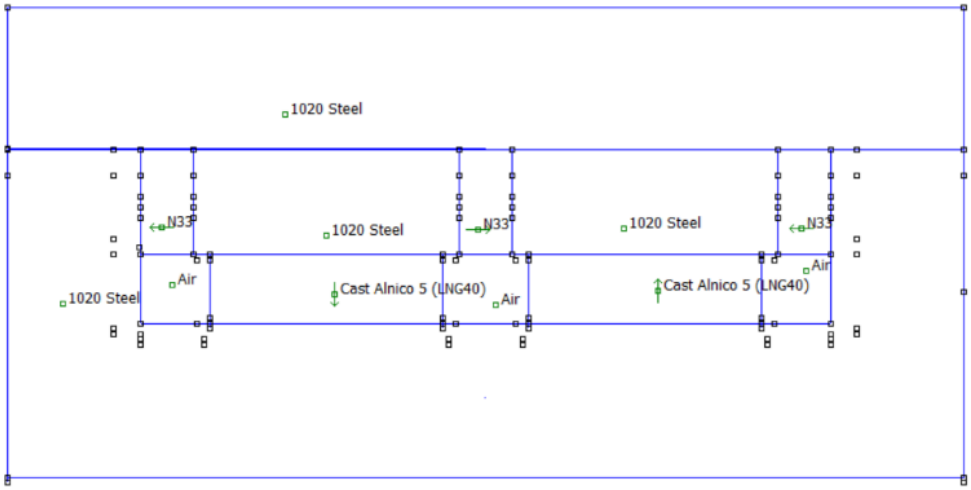
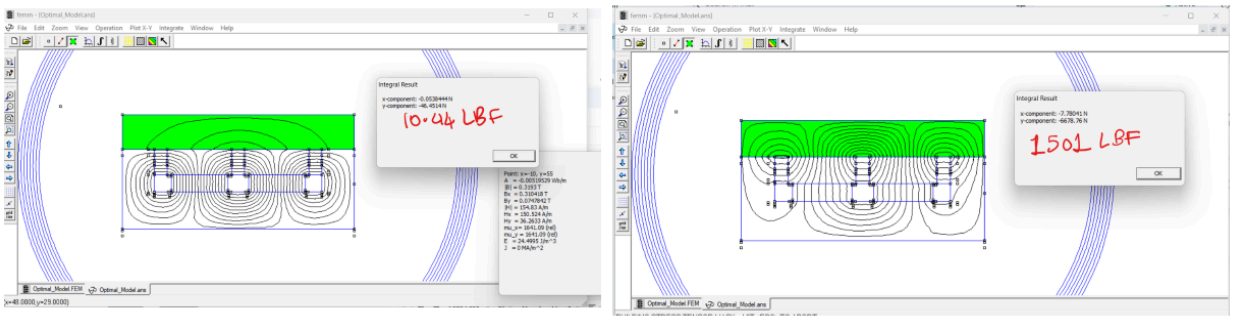


Figure 3 : Model achieved by brute-force approach



OFF-STATE **ON-STATE**

Figure 4 : Brute-Force approach outputs

OctaveFEMM Analysis

We combined brute-force testing with OctaveFEMM, a MATLAB toolbox for automated FEMM simulations. The initial design was iteratively refined (Fig. 2) using programmed changes and saved results for efficient analysis. Project constraints dictated dimensions like magnet widths, pole width, and bobbin size. We manually increased pole height to 25mm for lower off-state force and flux control. The workpiece-EPMC gap was set at 0.3mm. Based on these constraints, the NdFeB width was calculated as 10mm.

A MATLAB code generated the FEMM model. We aligned the top edges of NdFeB magnets with the steel poles and incrementally lowered their bottoms (0.5mm steps) while maintaining grade and calculating force integral via stress tensor.

Fringing flux necessitated increasing the case size to 180mm x 65mm, requiring a program rerun. OctaveFEMM analysis showed increasing AlNiCo magnet size raised the ON-state force, while increasing NdFeB size lowered the OFF-state force. Results are documented in Appendix A.

EPMC activation is simulated by switching AlNiCo magnet polarity in FEMM.

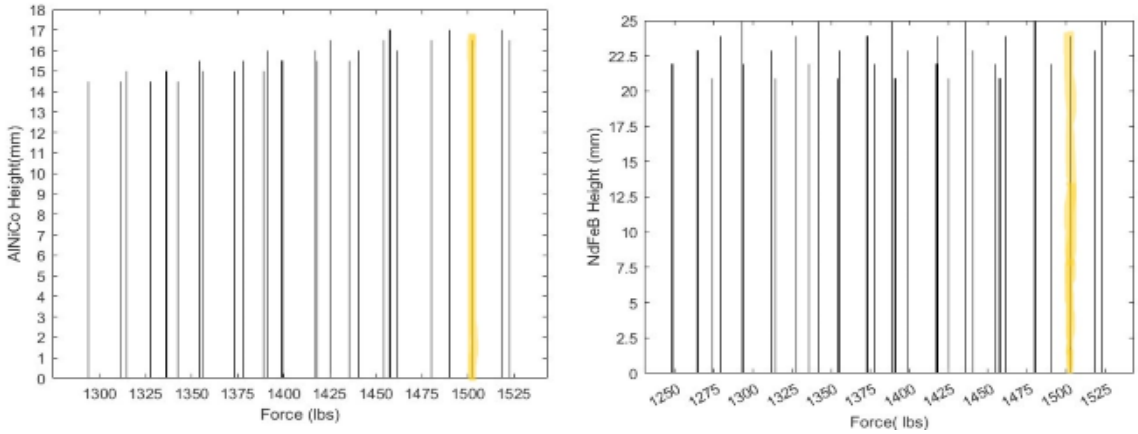


Figure 5 : ON State forces

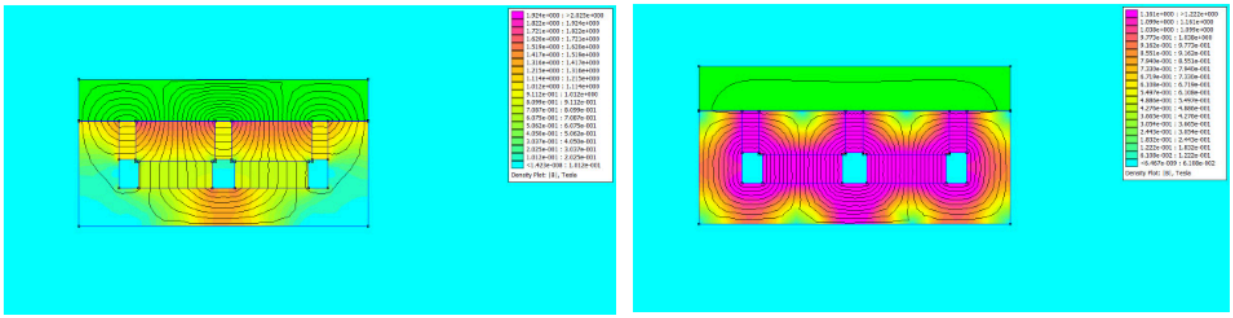
The graphs above illustrate the range of force variations concerning the dimensions of the magnets (AlNiCo and NdFeB). These observations were conducted using N33 and LNG 40, which were identified as highly suitable grades based on their B-H characteristic curves as shown in Appendix B. Additionally, we explored various

combinations of magnet grades to analyze the most cost-effective solution, ultimately selecting the aforementioned magnet grades.

Upon obtaining the results, the optimal dimensions for the magnets, where we achieved a maximum force of 1501 lbf and minimum force of 10.44 lbf during the off state, as shown in Table 2.

AlNiCo Height (ON & OFF state)	NdFeB height (ON & OFF state)	Force (lbf)	
		On State	Off State
16.5mm	23.9mm	1501	10.44

Table 2 : Results of OctaveFEMM based approach



ON-STATE **OFF-STATE**

Figure 6 : Outputs of MATLAB code

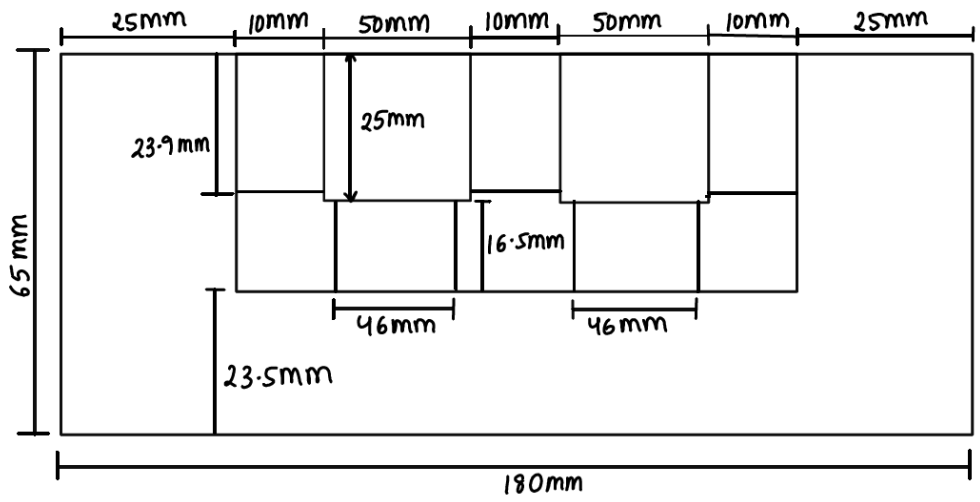


Figure 7 : Final Dimensions of Electro-Permanent Magnet Clamp

Thermal Analysis

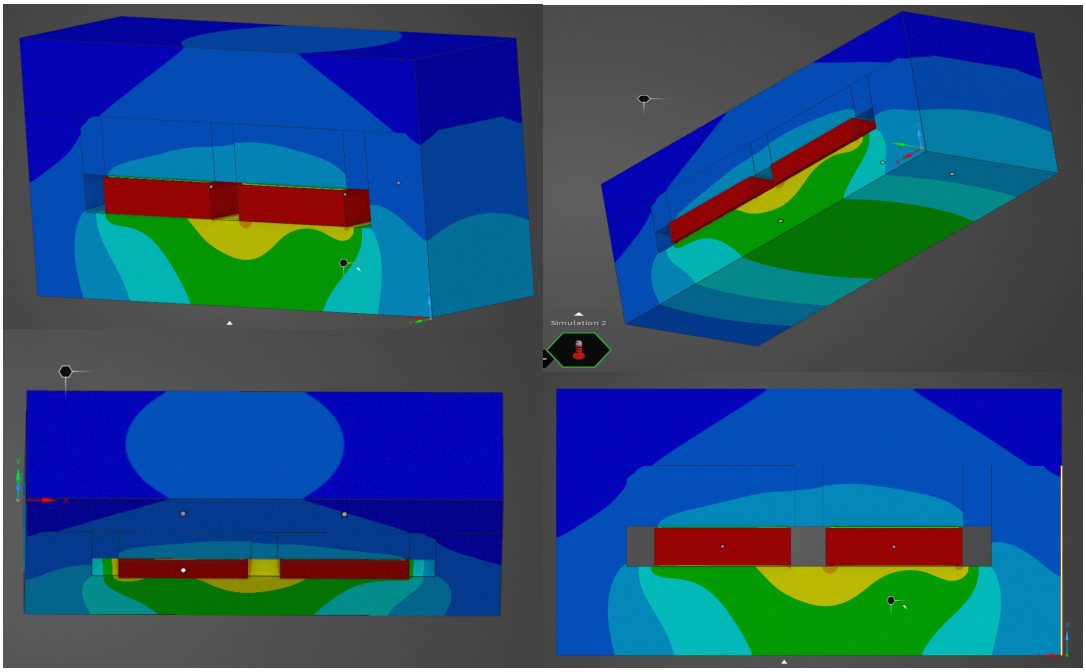


Figure 8 : Thermal analysis showing Top, Bottom and Front View of the model

The thermal analysis was performed in ANSYS Discovery to test whether the temperature increase due to the current through the windings of copper wire did not pass the threshold of the copper and damage the copper inside. Furthermore we wanted to know if it will also affect the magnets and its surroundings.

The system has an input of 230V AC which is then connected to a step down transformer that lowers the voltage to 80V AC. After it is rectified, filtered and used as the input voltage for a converter that is controllable and can pulsate the current through the system depending on the duty cycle. It can go from 0A "OFF" state 40A "ON" state max and change direction.

The copper cables chosen for this project are the AWG 24 and by using pulse current through them you can achieve higher currents and maintain temperatures. The copper was simulated using the dimensions given in the bobbin document as part of the problem and is a solid square piece surrounding the Alnico permanent magnet. In Ansys the temperature threshold of the copper and the other material that were used in the design were added. Then the test was run to get to the maximum temperature generated on the components at the maximum value of the current (40A pulsating) and its effects on the surrounding materials were observed.

The test was run with a setting to get to steady state condition resulting in no / negligible temperature change per second. The maximum temperature seen at steady state was 45 ℃ and this was around the copper cables which has a threshold temperature of 150 ℃. All other temperatures Alnico, NdFeB and steel stayed below their maximum.

The results of this simulation were a little surprising since it showed that a lot of the heat in our design went down towards the base and not to the workpiece even though it was at maximum current. This means that when you take the workpiece out of the machine the temperature will be relatively stable or if it gets hot it will not be hazardous to the touch for the person handling the workpiece.

Cost Analysis

In the cost analysis we have mainly considered the 3 major components that are crucial to the overall pricing. Based on our design, it is safe to say that using lower volumes of lower grade NdFeB magnets and higher grade of AlNiCo magnets is a better option, since N50 or N52 can be quite expensive. Furthermore, there is a tradeoff between the different gauges of copper wires, required length and cost. Higher gauges would reduce the number of turns required but the cost per spool would be very high. [References mentioned in appendix A].

Material	Total Weight/Length	Approximated Cost ($)
AlNiCo (LNG40)	0.72 Kg	3.83
NdFeB (N33)	0.2018 Kg	3.69
Copper wire(AWG 24)	56 m	11.24
Total Cost		≅ **$18.7**

Table 3 : Cost Analysis

Conclusion

A three-stage approach was employed to analyze the EPMC, combining manual calculations and simulation-aided methods. This was followed by a thermal analysis of the developed model. The analysis revealed a critical trade-off between the grade and volume of the magnet materials used. Additionally, reducing the air gap played a significant role in achieving the desired outcomes. Furthermore, increasing the case dimensions proportionally improved flux captured within the EPMC during the OFF-state and facilitated a smooth transition of flux into the workpiece during the ON-state. The design utilized AWG 24 magnet wire with a controllable pulse current ranging from 0 A (OFF state) to 40 A (maximum ON state). The thermal analysis identified a temperature of 45 ℃ on the copper cables using the specified circuit. This temperature can be further optimized through adjustments to the cable gauge.

Appendix A :

1. Lua language Reference Manual: https://www.lua.org/docs.html
2. Magnet Manufacturers USA: https://www.zhaobao-magnet.com/top-magnet-manufacturers-in-usa/
3. Magnet Pricing and Raw Material: https://allianceorg.com/wp-content/uploads/2019/07/Magnet-and-Raw-Material-Prices.pdf
4. Neodymium magnets with drastic price developments: https://www.magnosphere.co.uk/information-permanent-magnet/price-explosion-for-neodymium-magnets-raw-material/
5. Neodymium magnets details and Pricing: https://www.metal.com/Rare-Earths-Magnets/202103120022
6. Alnico magnets Pricing: https://magnet.com.au/products/alnico-block-magnet-50mm-x-15mm-x-10mm
7. Alnico magnets Pricing: https://zjzdgt.en.made-in-china.com/product/LxwYsyAVATWO/China-AISI-1020-Carbon-Steel-Flat-Iron-Metal-Flat-Bar-Carbon-6mm-Flat-Steel-Price.html
8. Density Calculator: https://www.calculator.net/density-calculator.html?cdensity=7.3&cdensityunit=gram%2Fcubic+centimeter&cvolume=9.9e-5&cvolumeunit=cubic+meter&cmass=&cmassunit=kilogram&ctype=&x=Calculate
9. Fermilab 1020 B vs H Curve: https://inis.iaea.org/collection/NCLCollectionStore/_Public/16/053/16053329.pdf?r=1
10. BH Curve for Neodymium Magnets Pricing: https://www.kjmagnetics.com/bhcurves.asp?gad_source=1&gclid=CjwKCAjww_iwBhApEiwAuG6ccP1uE24xNVeJCMqMxnqfQ72ar_r_kHGtI_iB2xcRCKLzYS5YeW8-kBoCQ_UQAvD_BwE
11. CHISEN: AISI 1020 Carbon steel: https://www.forgedproduct.com/forging-materials/astm-sae-aisi-1025-carbon-steel.html#:~:text=AISI%201025%20Carbon%20Steel%20
12. Goudsmit AlNiCo Grade System: https://www.goudsmit.co.uk/app/uploads/2023/05/AlNiCo-grades-available-at-Goudsmit.pdf
13. Cable sizing: https://diy.stackexchange.com/questions/195130/does-the-nec-have-a-table-for-oversizing-wire-for-extended-lengths
14. Copper wire pricing: https://www.digikey.com/en/products/detail/cnc-tech/600224/4693812

Appendix B:

ON State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N35, AlNico: LNG40					
NdFeb Height / Alnico Height	20.9mm	21.9mm	22.9mm	23.9mm	24.9mm
14mm	1231.019619	1248.786137	1264.885016	1279.683121	1293.562579
14.5mm	1274.324103	1293.882898	1311.567272	1327.64948	1342.354827
15mm	1314.710333	1336.152981	1355.539154	1373.024203	1388.90469
15.5mm	1354.097528	1377.944715	1399.11465	1418.130918	1435.392469
16mm	1390.853015	1417.319079	1440.606927	1461.321358	1480.014773
16.5mm	1425.037717	1454.311576	1479.942282	1502.497062	1522.763912
17mm	1457.526783	1490.022944	1518.395779	1543.170133	1544.170133

Table 4. ON State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N35 Alnico: LNG40

OFF State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N35, AlNico: LNG40					
NdFeb Height / Alnico Height	20.9mm	21.9mm	22.9mm	23.9mm	24.9mm
14mm	6.024829546	3.123457501	1.86120758	2.242665661	4.255866261
14.5mm	6.041907237	3.137709106	1.878455063	2.270835565	4.305406897
15mm	6.054188134	3.139650193	1.884513022	2.285196341	4.338922276
15.5mm	6.05942876	3.143908075	1.893023235	2.307749308	4.38709456
16mm	6.058108008	3.155588031	1.904904279	2.335151905	4.431674807
16.5mm	6.042509538	3.13834467	1.902174112	2.343218752	4.47392102
17mm	6.019088625	3.112341062	1.896975529	2.354882546	4.507320118

Table 5. OFF State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N35 Alnico: LNG40

ON State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N30, AlNico: LNG40					
NdFeb Height / Alnico Height	20.9mm	21.9mm	22.9mm	23.9mm	24.9mm
14mm	1196.421951	1215.640339	1233.019596	1248.846691	1263.358239
14.5mm	1235.917826	1257.458216	1276.707158	1294.150156	1309.916138
15mm	1271.986213	1296.103972	1317.472115	1336.617511	1353.91398
15.5mm	1306.413103	1333.528914	1357.403249	1378.471675	1397.470328
16mm	1338.179228	1368.106776	1394.791687	1418.175836	1438.954401
16.5mm	1367.599051	1400.130201	1429.557474	1455.489384	1478.38246
17mm	1395.407235	1430.758458	1462.878205	1491.61964	1517.013573

Table 6. ON State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N30 Alnico: LNG40

OFF State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N30, AlNico: LNG40					
NdFeb Height / Alnico Height	20.9mm	21.9mm	22.9mm	23.9mm	24.9mm
14mm	14.03696834	8.871561884	5.099217962	2.722161893	1.734554999
14.5mm	14.07865156	8.899889464	5.118432745	2.739291215	1.75560626
15mm	14.11015232	8.91087719	5.121647596	2.739060423	1.760271626
15.5mm	14.13458359	8.923328544	5.126667745	2.744336964	1.775618591
16mm	14.15231479	8.935699019	5.130820187	2.750535804	1.787033948
16.5mm	14.15238625	8.920950698	5.116101257	2.735011644	1.787578959
17mm	14.14178229	8.89852443	5.094230547	2.717822822	1.777706871

Table 7. OFF State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N30 Alnico: LNG40

ON State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N33, AlNico: LNG40					
NdFeb Height / Alnico Height	20.9mm	21.9mm	22.9mm	23.9mm	24.9mm
14mm	1218.114139	1236.408686	1252.970377	1268.097435	1282.119183
14.5mm	1260.102693	1280.313869	1298.5715	1315.134637	1330.164668
15mm	1298.992469	1321.31778	1341.374928	1359.469159	1375.835879
15.5mm	1336.550602	1361.618724	1383.661346	1403.396487	1421.288538
16mm	1371.377879	1399.271069	1423.762791	1445.339801	1464.781553
16.5mm	1403.670142	1434.379586	1461.507696	1485.195211	1506.336374
17mm	1434.27049	1468.096655	1498.062507	1524.326743	1547.463626

Table 8. ON State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N33 Alnico: LNG40

OFF State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N33, AlNico: LNG40					
NdFeb Height / Alnico Height	20.9mm	21.9mm	22.9mm	23.9mm	24.9mm
14mm	8.650540891	4.814654501	2.519152997	1.767531644	2.550663313
14.5mm	8.67485786	4.831925811	2.534437548	1.788442228	2.585729701
15mm	8.692161945	4.834893692	2.536744052	1.794232499	2.604760814
15.5mm	8.7024905	4.839800522	2.541198952	1.807158385	2.636797949
16mm	8.706236704	4.849084974	2.547344313	1.823360536	2.665323579
16.5mm	8.693835391	4.830022796	2.537087212	1.819277287	2.687848155
17mm	8.672092854	4.802506872	2.522326166	1.816501967	2.700964233

Table 9. OFF State Force (lbf) Mapping at different dimensions with Material properties NdFeb: N33 Alnico: LNG40

Force Equations:

1: $R_{air}\mathbf{\Phi}_1 + R_{steel} + R_{air}\mathbf{\Phi}_2 + R_{NdFeB}\mathbf{\Phi}_3 = 0$

2: $R_{steel}\mathbf{\Phi}_4 + R_{air}\mathbf{\Phi}_2 + R_{NdFeB}\mathbf{\Phi}_5 + R_{air}\mathbf{\Phi}_2 = 0$

3: $-R_{NdFeB}\mathbf{\Phi}_3 - R_{Alnico}\mathbf{\Phi}_6 + R_{steel-base}\mathbf{\Phi}_7 = 0$

4: $R_{NdFeB}\mathbf{\Phi}_5 + R_{Alnico}\mathbf{\Phi}_6 - R_{steel-base}\mathbf{\Phi}_8 + R_{Alnico}\mathbf{\Phi}_6 = 0$

Node Equations:

5: $\mathbf{\Phi}_1 - \mathbf{\Phi}_2 + \mathbf{\Phi}_4 = 0$

6: $\mathbf{\Phi}_2 - \mathbf{\Phi}_3 - \mathbf{\Phi}_5 - \mathbf{\Phi}_6 = 0$

7: $\mathbf{\Phi}_6 - \mathbf{\Phi}_7 - \mathbf{\Phi}_8 = 0$

7: $\mathbf{\Phi}_7 - \mathbf{\Phi}_1 + \mathbf{\Phi}_3 = 0$

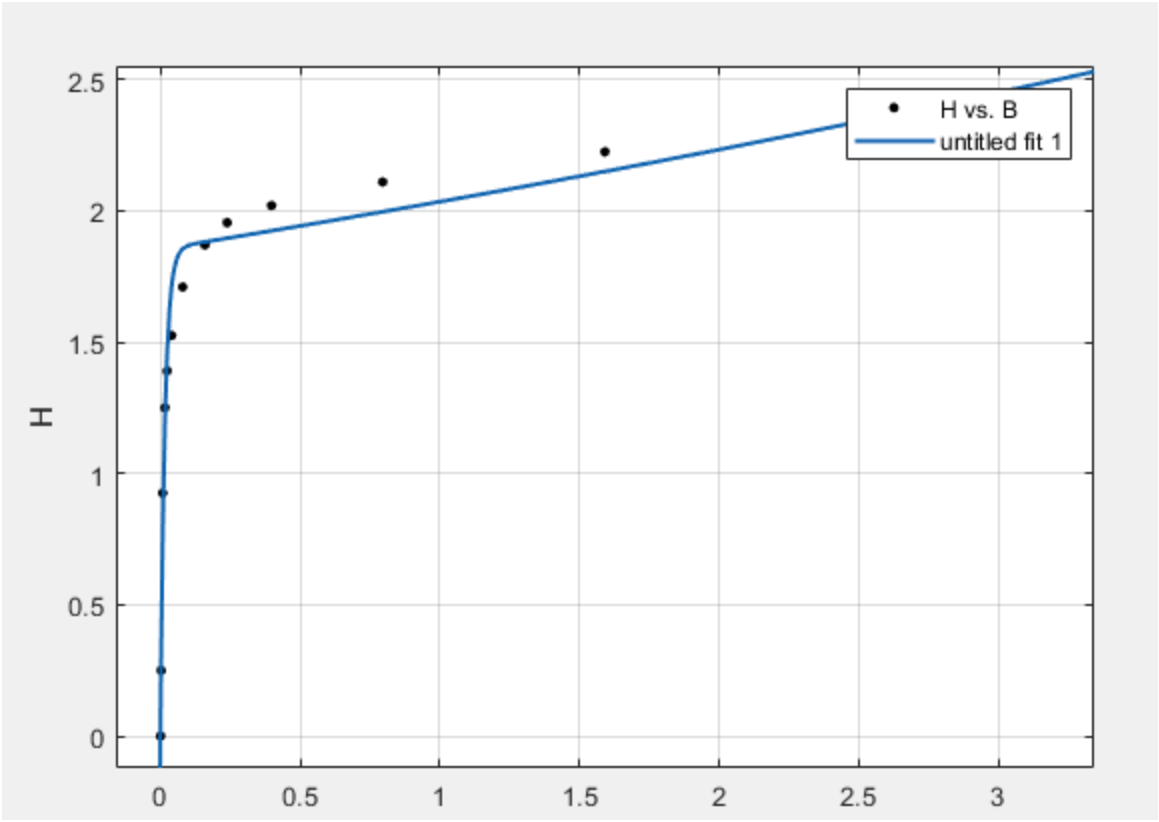


Figure. 9 B-H Steel 1020 Curve Fitting on matlab

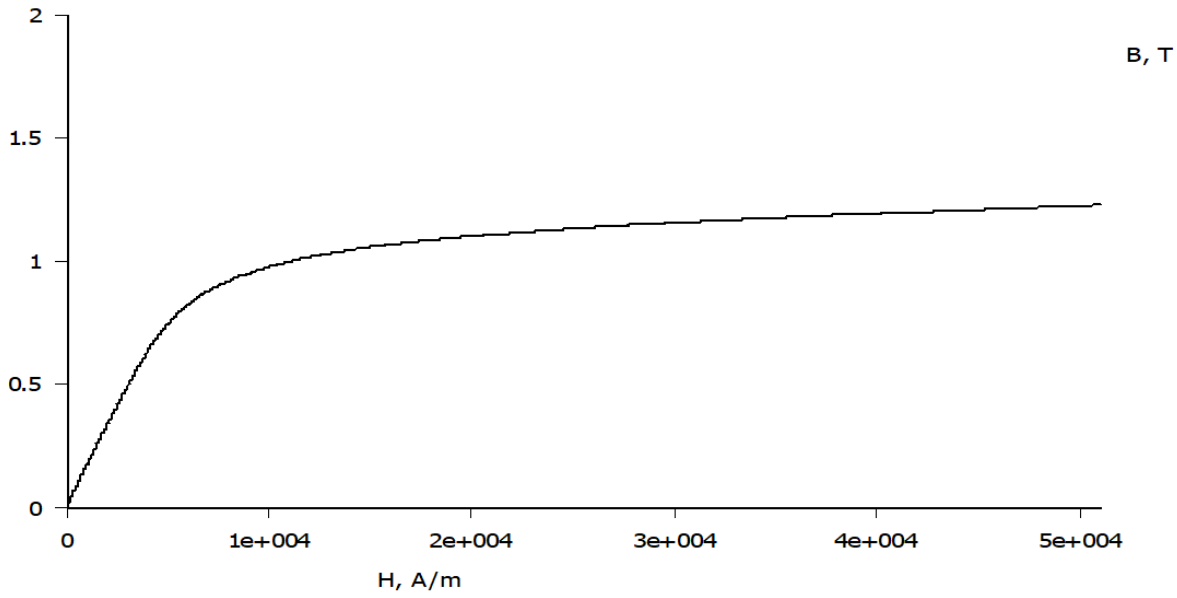


Figure.10 B-H Curve for Alnico from FEMM Plot tab

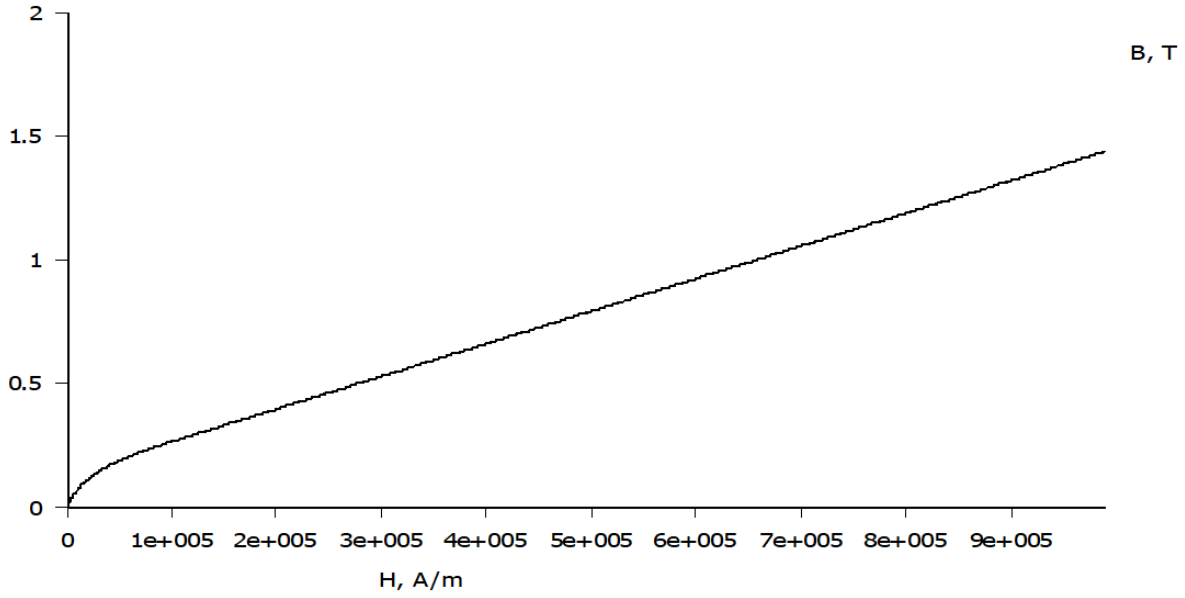


Figure 11. B-H Curve for NdFeB from FEMM Plot tab